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Higher Categories II:

More origins & the basics

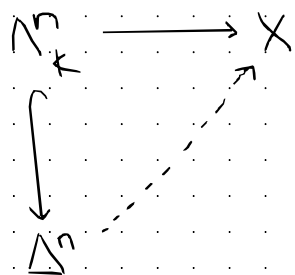
Last time Simplicial Sets, Kan complexes, the Dold-Kan theorem

Today More homotopy theory background

Recall For $0 \leq k \leq n$, the k -th horn in $\Delta^n \in \mathbf{sSet}$ is

$$\Lambda_k^n = \bigcup_{i \in [n] - \{k\}} \Delta^{[n] - \{i\}} \longleftarrow \text{face across from } k \text{ removed}$$

Def. A Simplicial Set X is a Kan complex if every horn filling problem



for $n \geq 1$ and
 $0 \leq k \leq n$

has a solution. Equivalently, the restriction map

$$X_n = \text{Hom}_{\text{sSet}}(\Delta^n, X) \longrightarrow \text{Hom}_{\text{sSet}}(\Lambda_K^n, X)$$

is a surjection.

Idea. There is a weakly-defined composition of edges

$$\begin{array}{ccc} & \nearrow g & \\ \bullet & \xrightarrow{f} & \bullet \end{array} \quad \rightsquigarrow \text{a map } \Lambda_1^2 \longrightarrow X$$

$$g \cdot f \quad \rightsquigarrow \text{a choice of filler } \Delta^2 \longrightarrow X$$

Ex. Every simplicial group is a Kan complex. If T is any topological space, then $\text{Sing}(T)$ is a Kan complex.

The homotopy hypothesis

Def A map of simplicial sets $f: X \rightarrow Y$ is a weak homotopy equivalence if $|f|: |X| \rightarrow |Y|$ is a weak homotopy equivalence of topological spaces.

Remark There is a completely intrinsic definition that doesn't use topological spaces, but it would take us some time to formulate

Thm (Kan, Quillen) The adjunction

$$|-|: \mathbf{sSet} \rightleftarrows \mathbf{Top} \text{ Sing}$$

induces an equivalence of categories

$$\mathbf{sSet}[\text{whe}^{-1}] \simeq \mathbf{Top}[\text{whe}^{-1}]$$

Moreover, write $\mathbf{Kan} \subset \mathbf{sSet}$ for the full subcategory spanned by the Kan complexes and $\mathbf{CW} \subset \mathbf{Top}$ for the full

Subcategory spanned by the spaces that admit CW structures. Then there are equivalences

$$\text{Kan}[\text{he}^{-1}] \simeq \text{sSet}[\text{whe}^{-1}] \quad \text{and} \quad \text{CW}[\text{he}^{-1}] \simeq \text{Top}[\text{whe}^{-1}]$$

homotopy equivalences

Upshot. The homotopy theory of spaces is 'the same' as the homotopy theory of Kan complexes/ ∞ -groupoids.

Remark. One might have the idea to 'do homotopy theory' by trying to work in the category

$$\text{sSet}[\text{whe}^{-1}] \simeq \text{Top}[\text{whe}^{-1}]$$

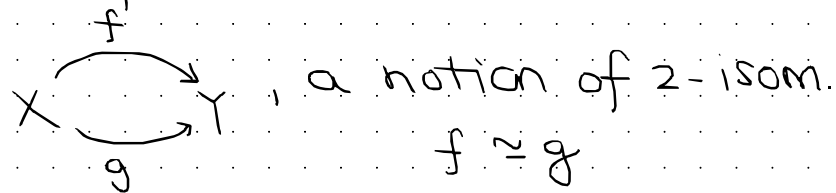
This does not work. The problem is that this homotopy category of spaces has almost no limits/colimits. The underlying problem is that we're only remembering that two maps are homotopic, but not how they are homotopic.

Desire, have an enhanced notion of a category called an ∞ -category with

(0) Objects

(1) For two objects X, Y 1-morphisms $X \rightarrow Y$

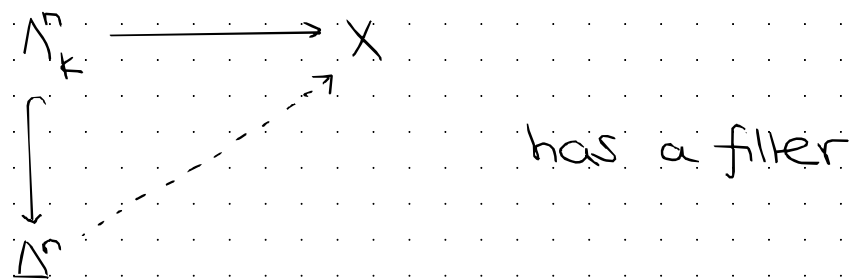
(2) For two 1-morphisms



Said differently, for objects X, Y , we should have a space/
Kan complex $\text{Map}(X, Y)$ of maps $X \rightarrow Y$ behaving 'as
expected'

> One way to make this precise is to weaken Grothendieck's
description of the nerve of a category

Def. A simplicial set is a quasicategory or ∞ -category if for each $n \geq 2$ and $0 < k < n$, each inner horn



Ex. If $\mathcal{C}$ is a category, then the simplicial set $N(\mathcal{C})$ is a quasicategory.

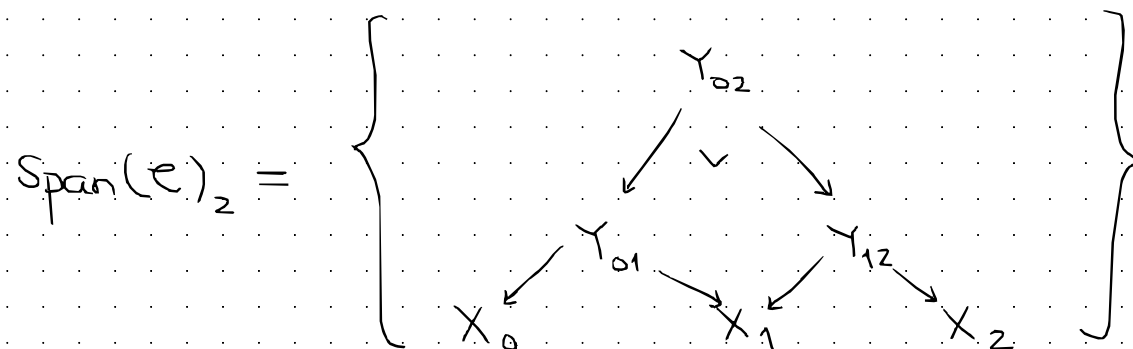
Ex. Let $\mathcal{C}$ be a category with pullbacks. Define a simplicial set $\text{Span}(\mathcal{C})$ with n -simplices:

(0) $\text{Span}(\mathcal{C})_0 = \text{obj}(\mathcal{C})$.

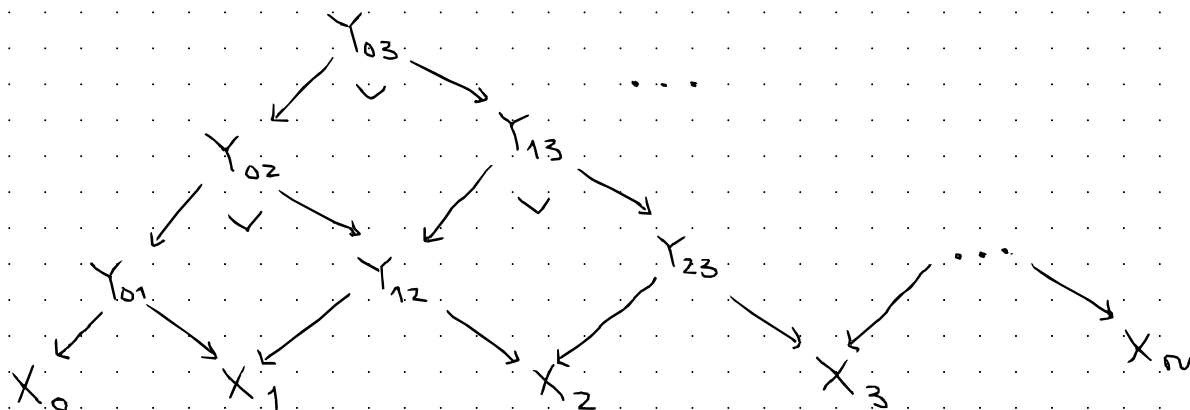
(1) $\text{Span}(\mathcal{C})_1 = \left\{ \text{diagrams } \begin{array}{ccc} & Y & \\ X_0 \swarrow & & \searrow \\ & X_1 & \end{array} \right\}$

(2)

$\text{Span}(e)_2 =$



(n) $\text{Span}(e)_n$ is the set of diagrams



> The face and degeneracy maps are given by composing or adding identities.

> Check that this simplicial set is a quasicategory.

Remark This simplicial set is one way to encode the $(2,1)$ -category with morphisms spans in $\mathcal{C}$ and composition defined by pullback.

Construction (Internal Hom in $\mathbf{sSet}$). Let X and Y be simplicial sets. Write $\mathrm{Fun}(X, Y)$ for the simplicial set

$$\mathrm{Fun}(X, Y)_n = \mathrm{Hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$$

One can easily check that $\mathrm{Fun}(X, Y)$ is the internal Hom in $\mathbf{sSet}$.

Ex. If $\mathcal{C}$ and $\mathcal{D}$ are categories, there is a natural iso

$$N(\mathrm{Fun}(\mathcal{C}, \mathcal{D})) \xrightarrow{\sim} \mathrm{Fun}(N(\mathcal{C}), N(\mathcal{D}))$$

Lemma. If X and Y are simplicial sets and Y is a quasi-category, then $\text{Fun}(X, Y)$ is a quasi-category.

Loyal's Characterization of quascategories

Def. Let $n \geq 0$. The boundary of Δ^n is the simplicial subset

$$\partial\Delta^n = \bigcup_{i \in [n]} \Delta^{[n] \setminus \{i\}} \quad \leftarrow \text{only remove the } n\text{-simplex} \\ \text{corr to } \text{id}: [n] \longrightarrow [n]$$

A map of simplicial sets $f: X \rightarrow Y$ is a trivial Kan fibration if every solid diagram

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dotted} & \downarrow f \\ \Delta^n & \xrightarrow{\quad} & Y \end{array}$$

has a dotted filler

> This guarantees all the fibers of f are contractible Kan complexes.

Thm (Joyal) A simplicial set X is a quascategory if and only if the restriction map

$$\mathrm{Fun}(\Delta^2, X) \longrightarrow \mathrm{Fun}(\Lambda_1^2, X)$$

is a trivial Kan fibration of simplicial sets.

> So, in particular, for every composable pair of arrows $\Lambda_1^2 \rightarrow X$, there is a contractible Kan complex of possible compositions.

History Joyal and Lurie have thoroughly developed ∞ -category theory from this perspective, and verified that all of the 'expected' theorems hold.

> However, this was a highly nontrivial task!

> The key point is that how many results/definitions in category theory are typically formulated, they may not immediately adapt well to this setting. But there is a formulation that

does adapt well.

Example Let X be an ∞ -category and $x, y \in X_0$ objects. We write $\text{Map}_X(x, y)$ for the pullback of simplicial sets

$$\begin{array}{ccc} \text{Map}_X(x, y) & \longrightarrow & \text{Fun}(\Delta^1, X) \\ \downarrow \wr & & \downarrow \text{restrict} \\ \Delta^0 & \xrightarrow{(x, y)} & X \times X \cong \text{Fun}(\partial\Delta^1, X) \end{array}$$

One has to check that $\text{Map}_X(x, y)$ is a Kan complex. Warning In higher category theory, it is typical to write Map instead of Hom .

Example Once we have formulated mapping Kan complexes, one can then talk about initial/terminal objects. For example, $y \in X_0$ is terminal if for every $x \in X_0$, the Kan complex $\text{Map}_X(x, y)$ is contractible.

> To talk about (co)limits, we need to be able to adjoin a new initial/terminal object to any ∞ -category

Construction (join). Write $(-)*(-): \Delta \times \Delta \rightarrow \Delta$ for the nonunital monoidal structure with

$I * J :=$ the set $I \sqcup J$ ordered so that

$I \hookrightarrow I \sqcup J$ and $J \hookrightarrow I \sqcup J$ are
order preserving and all
 $i \in I$ are smaller than all $j \in J$.

So if $I = \{i_0 < \dots < i_m\}$ and $J = \{j_0 < \dots < j_n\}$, then

$$I * J = \{i_0 < \dots < i_m < j_0 < \dots < j_n\}$$

That is, $[m] * [n] \cong [m+n+1]$. Define a monoidal operation on simplicial sets by left Kan extension

$$\begin{array}{ccc}
 \Delta \times \Delta & \xrightarrow{\quad * \quad} & \Delta \\
 \downarrow \text{J} \times \text{J} & & \downarrow \text{J} \\
 \text{sSet} \times \text{sSet} & \xrightarrow[\quad * \quad]{} & \text{sSet}
 \end{array}$$

That is, $*$ is the unique operation that preserves colimits separately in each variable and is given on representables by

$$\Delta^I * \Delta^J = \Delta^{I * J}$$

Def. Given a quasicategory X , we write

$$X^\triangleleft = \Delta^0 * X \quad \text{and} \quad X^\triangleright = X * \Delta^0$$

So $X^\triangleleft$ is X with a new initial object, and $X^\triangleright$ is X with a new terminal object, we call these new objects $-\infty$ and $+\infty$, respectively.

Def Let $p: I \rightarrow X$ be a map between quascategories. The quascategory of Cones over p is the pullback

$$\mathcal{C}_{/p} := \text{Fun}(I^{\Delta}, X) \times_{\text{Fun}(I, X)} \{p\}$$

We say that a limit of p exists if $\mathcal{C}_{/p}$ has a terminal object $\bar{p}: I^{\Delta} \rightarrow X$. In this case, we call $\bar{p}(-\infty) \in X_0$ the limit of the diagram p .

Upshot As we hope these examples demonstrate, once one has some basic constructions set up, we can formulate definitions, results, etc in a way that doesn't refer to a specific definition of the term ' ∞ -category'.

> See Chapter 1 of Crissen's 'stable homotopy theory and higher algebra' for how to axiomatically set up this 'model agnostic' approach.

Axiomatics

> Even though we won't go through the axiomatics in detail, we'll start using these tools and hopefully demonstrate how to work with the axioms.

Def An ∞ -category $\mathcal{C}$ is an anima (or ∞ -groupoid or space) if every map in $\mathcal{C}$ is invertible, i.e., for every map $f: x \rightarrow y$, there exist maps $g: y \rightarrow x$ and $h: y \rightarrow x$ along with equivalences $gf \simeq \text{id}_x$ and $fh \simeq \text{id}_y$. We write $\text{An} \hookrightarrow \text{Cat}_\infty$ for the full subcategory spanned by the anima.

Axiom For each ∞ -category $\mathcal{C}$, there exists a natural functor

$$\text{Map}_{\mathcal{C}}(-, -): \mathcal{C}^{\text{op}} \times \mathcal{C} \longrightarrow \text{An}.$$

Moreover, this functor preserves colimits separately in each variable.

Axiom. There are functors $s\text{Set} \rightarrow \text{An}$ and $\text{Top} \rightarrow \text{An}$ that exhibit An as the localizations (in the sense of ∞ -categories)

$$\text{An} \simeq s\text{Set}[\text{whe}^{-1}] \simeq \text{Top}[\text{whe}^{-1}].$$

Axiom. The inclusion $\text{An} \hookrightarrow \text{Cat}_{\infty}$ admits both a left and a right adjoint (hence An has all limits and colimits). The left adjoint sends an ∞ -category $\mathcal{C}$ to its classifying anima

$$B\mathcal{C} = \tau[\mathcal{C}^{-1}]$$

The right adjoint sends $\mathcal{C}$ to its maximal sub-anima $\mathcal{C}^{\simeq}$

Notation. For an ∞ -category $\mathcal{C}$, write $\text{Psh}(\mathcal{C}) := \text{Fun}(\mathcal{C}^{\text{op}}, \text{An})$. and write $\mathcal{C} \rightarrow \text{Psh}(\mathcal{C})$ for the functor $c \mapsto \text{Map}_{\mathcal{C}}(-, c)$.

Yoneda Lemma. Let $\mathcal{C}$ be a small ∞ -category. For any $c \in \mathcal{C}$ and $F \in \text{Psh}(\mathcal{C})$, there is a natural equivalence

$$\text{Map}_{\text{Psh}(\mathcal{C})}(\mathcal{Y}(c), F) \xrightarrow{\sim} F(c)$$

Moreover, the functor $\mathcal{Y}: \mathcal{C} \rightarrow \text{Psh}(\mathcal{C})$ is fully faithful.

Universal Property of presheaves Let $\mathcal{C}$ be a small ∞ -category and let $\mathcal{D}$ be an ∞ -category with colimits. Then restriction along the Yoneda embedding defines an equivalence of ∞ -categories

$$\text{Fun}^{\text{colim}}(\text{Psh}(\mathcal{C}), \mathcal{D}) \xrightarrow{\sim} \text{Fun}(\mathcal{C}, \mathcal{D})$$

colimit-preserving functors

$$[\bar{F}: \text{Psh}(\mathcal{C}) \rightarrow \mathcal{D}] \longmapsto [\bar{F} \circ \mathcal{Y}: \mathcal{C} \rightarrow \mathcal{D}]$$

The inverse is given by sending a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ to its left Kan extension along the Yoneda embedding

$$\bar{F}(X) := \text{colim}_{\mathcal{Y}(c) \rightarrow X} F(c)$$

Moreover, every colimit-preserving functor $\bar{F}: \text{PSh}(\mathcal{C}) \rightarrow \mathcal{D}$ is a left adjoint with right adjoint $G: \mathcal{D} \rightarrow \text{PSh}(\mathcal{C})$ given by

$$G(d)(c) = \text{Map}_{\mathcal{D}}(F(c), d).$$